

Parthiv Gajula

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Professional Summary

Focused on building large language models and backend systems, with hands-on experience training and modifying models while studying behavior in real-world environments. Built end-to-end systems in Agile development environments. 2× Hackathon Winner.

Professional Experience

365 Retail Markets

May 2025 – Aug 2025

Database Intern

- Optimized SQL queries, improving execution speed by up to 45% through indexing and execution plan analysis
- Configured database replication across remote instances to ensure high availability in distributed systems
- Developed SQL triggers to automate data processing and enforce integrity constraints

Projects

TraceLens – Distributed Tracing Platform (Hackathon)

March 2026

- Built an observability platform using OpenTelemetry and Jaeger to trace microservice request flows
- Deployed services in Docker containers for real-time trace ingestion and analysis
- Engineered a NestJS pipeline to reconstruct cross-service request paths and span relationships
- Developed a graph-based UI to visualize execution flows and identify latency bottlenecks

SpeechMatch – Capstone Team Lead

Jan 2026 – Present

- Leading development of a mobile pronunciation platform with an end-to-end speech evaluation pipeline
- Built backend services for audio processing, OCR ingestion, and scoring using Azure Speech Services
- Implemented timing and deviation scoring using word-level alignment and edit distance

Retrieval-Augmented Generation (RAG) Research Study

Jan 2025 – May 2025

- Implemented RAG pipelines using DPR, ColBERT, and SELF-RAG architectures
- Built retrieval systems with document chunking, indexing, and embedding-based search
- Benchmarked RAG vs LLM-only generation using MS MARCO

HarmonAize – Heart Rate Music Recommendation (GrizzHacks Winner)

2024

- Built a music recommendation system based on target heart rate using clustering
- Improved performance by replacing low-correlation supervised models with unsupervised clustering
- Integrated ML inference into a full-stack application with a user interface

AWS S3 Encrypted FUSE Filesystem

Sep 2025 – Dec 2025

- Built a FUSE3 filesystem that encrypts files before upload to an S3 bucket
- Intercepted file descriptor cache to insert write operations prior to upload
- Designed a SHA-256 hashing pipeline for data integrity

Technical Skills

Languages: Python, Java, JavaScript, SQL, C, C++, Kotlin, Ruby, R

Frameworks: React, Next.js, Flask, Ruby on Rails, PyTorch

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Firebase

DevOps/Cloud: Docker, Kubernetes, Terraform, Jenkins, Nginx

Tools: Postman, SSMS, Expo

AI/ML: Transformers, BERT, GPT, RAG, Fine-Tuning, RLHF, Multimodal Models

Education

Wayne State University

Aug 2022 – May 2026

B.S. in Computer Science, Concentration in Artificial Intelligence